



Rutschblock-scale snowpack stability derived from multiple quality-controlled SnowMicroPen measurements

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ABSTRACT

Stability prediction from SnowMicroPen (SMP) profiles would support avalanche forecasting operations, since objective stability information could be gathered more quickly than with standard tests, thereby allowing sampling at higher resolution and over larger spatial scales. Previous studies have related the snow properties derived from the SMP to observed snow properties at Rutschblock (RB) and compression test failure planes. The goals of this study are to show to what extent snowpack stability for artificial triggering, based on RB, can be derived from SMP measurements and how multiple measurements at the RB scale improve the results. Measurements at 36 different sites were used for the development of a classification scheme. Each site included a RB test, a manual profile, and 6 to 10 adjacent SMP measurements, for a total of 262 SMP profiles. A recently improved SMP theory was applied to estimate the micro-structural and mechanical properties of manually defined weak layers and slab layers. SMP signal quality control and different noise treatment methods were taken into consideration in the analysis. The best and most robust predictor of RB stability was the weak layer micro-scale strength. In combination with the SMP-estimated mean density of the slab layer, the total accuracy of predicting RB stability classes was 85% over the entire dataset, and 88% when signals with obvious signal dampening (11% of the dataset) were removed. The total accuracy increased when multiple SMP measurements at the RB scale were used to calculate the mean weak layer strength, when compared to using just one SMP measurement at a site. The analysis was robust to trends and offsets in the absolute SMP force, which was a frequent signal error. However, it was sensitive to dampened or disturbed SMP force micro variance. The sensitivity analysis also showed that the best predictor of instability, the weak layer micro-scale strength, was robust to the choice of SMP signal noise removal method.

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1. Introduction

Snowpack measurements and stability tests are, along with observations of recent avalanche activity and weather history, the basis for snowpack stability assessments in most avalanche forecasting operations. The Rutschblock test is currently the standard snowpack stability test for operational avalanche forecasting in Switzerland. The Rutschblock score is an index of snow stability for skier triggering.

Schweizer and Jamieson (2003, 2007), Schweizer et al. (2008) developed a stability classification method based on stability test scores and manual observations of failure interface properties. These studies show the significance of observed snow properties at failure interfaces with respect to snowpack stability.

The SnowMicroPen (SMP), a high-resolution penetrometer for snow, measures penetration resistance or snow hardness at the grain scale (Schneebeli and Johnson, 1998). A physical theory for characterizing snow properties from the SMP signal was first introduced by Johnson and Schneebeli (1999). The three basic micro-structural parameters are the structural element length (L), the deflection at rupture (δ) and the rupture force (f). From these, mechanical parameters can be derived. Sturm et al. (2004) improved the theory by solving δ more accurately and by replacing the assumption that each element occupies a cubical volume with sides of length L with a spherical volume with a diameter of L . Marshall (2006) added several improvements including an algorithm for accounting for simultaneous ruptures. Pielmeier and Schweizer (2007) applied the theory by Johnson and Schneebeli (1999) and Pielmeier et al. (2006) applied theories by Johnson and Schneebeli (1999) and Marshall (2006) in statistical approaches to predict the stability of Rutschblock failure interfaces from the SMP signal. The older version of the theory gave a classification accuracy of 65% for discriminating stable and unstable failure interfaces. The newer version

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yielded a classification accuracy of 70%. In a further statistical approach based on Johnson and Schneebeli (1999), Bellaire et al. (2009) predicted snowpack stability from SMP measurements with an accuracy of 75%. Besides the SMP-based stability classifications, Floyer and Jamieson (2009) have classified fracture character based on SABRE penetrometer profiles.

Marshall and Johnson (2009) have further improved the SMP theory. Verified by both extensive Monte-Carlo simulated and observed snow signals, it yielded a much higher accuracy when estimating the micro-structural and mechanical parameters than with previous versions of the theory. Applying this improved theory by Marshall and Johnson (2009) to a dataset comprised of snow profiles, Rutschblock tests and SMP measurements, we obtained more accurate and robust predictors of stability from the SMP signals. The effect of signal quality control, SMP signal noise filtering and weak layer thickness on the accuracy of the results is quantified. Furthermore, the variance of the means of the best classifier, the weak layer micro-scale strength, was analyzed to test the assumption that the variance within a site is smaller for unstable conditions than for stable conditions. Finally, we quantified the effect of an increased number of SMP measurements at the Rutschblock scale on the accuracy of the results with Monte-Carlo simulations.

2. Data and quality

2.1. Data

At 36 different sites, multiple SMP profiles were measured next to stability tests in dry snow conditions between December 2007 and March 2008 in the Swiss Canton of Graubünden (Fig. 1). All locations were chosen for the operational assessment of regional avalanche danger. Altitudes ranged between 2000 and 2600 m a.s.l. and the slope angles ranged between 30 and 40°. The majority of the profiles were taken on north facing slopes (NW–N–NE: 69%, SE–S–SW: 15%, E: 6%, W: 11%).

A combined profile consisted of one manual profile with at least one vertical (only for stratigraphic referencing) and one slope-perpendicular SMP measurement adjacent to it, as well as a Rutschblock test with up to seven SMP measurements at its perimeter and center. For the analysis, 262 SMP measurements were available. The 36 observed Rutschblock scores ranged from 1 to 7.

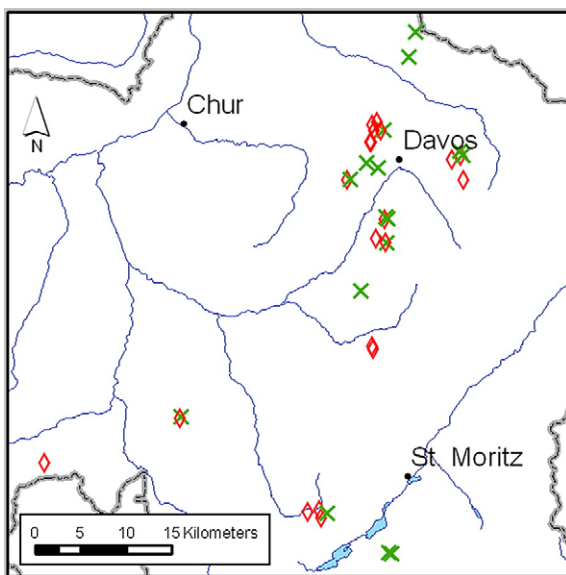


Fig. 1. Map of a portion of the Swiss Canton Graubünden showing the profile locations. Diamonds represent Rutschblock scores <4 and Rutschblock score 4 with release type whole block ($n=20$), crosses represent Rutschblock scores 4 with release type partial break and RB scores >4 ($n=16$).

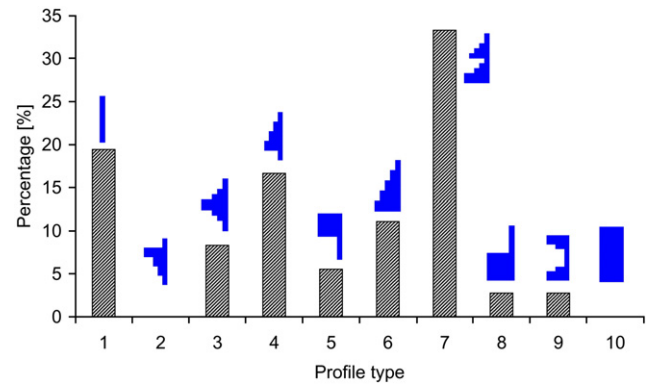


Fig. 2. The distribution of the 36 snow profiles in terms of profile types (icons). There is an equal split between profiles with weak (types 1–5) and consolidated basal layers (types 6–10).

Fig. 2 shows the range and frequency of snowpack type, classified following Schweizer and Lüscher (2001), at the 36 sites. The observed weak layers were in 80% of the sites persistent weak layers (40% small facets up to 1.5 mm, 40% large facets and depth hoar greater 1.5 mm) and in 20% of the sites new snow instabilities. The dataset contains only few very thin weak layers and no buried surface hoar layers. In the 262 SMP measurements only 5% of the weak layers are thinner than 5 mm.

2.2. SMP data quality

All 262 SMP measurements were qualitatively checked for obvious signal errors by considering the entire measurement. The profiles were grouped into four quality classes (Table 1). Class 1 (C1) data showed no error. Class 2 (C2) data showed a linear trend over 5 mm signal extend or an offset in the signal. Trends were classified artificial if the air signal showed a trend. Offsets were classified artificial relative to other measurements at the same site. Class 3 (C3) data showed a dampened signal micro-variance, where sections with lower variance than the normal air signal (± 0.02 N) exist. Class 4 (C4) data showed both errors, C2 and C3. Amongst the 36 sites were five sites that contained no C1 SMP measurements at all. Signal errors may stem from a variety of sources such as a frozen SMP measuring tip, a defective sensor or a defective cable. While the sources of error are not specified, the effect of data quality on the accuracy of the results is quantified. Lutz (2009) provides similar descriptions of SMP signal quality with examples and potential causes for poor quality signals.

3. Methods

3.1. Field methods

Manual snow profiles were taken according to the International Classification for Seasonal Snow on the Ground (Colbeck et al., 1990). Three different stability tests were performed adjacent to the manual profile: one Rutschblock test (RB), two extended column tests (ECT) and two compression tests (CT). Fig. 3 shows the experimental design

Table 1
Frequency and percentage of SMP profiles, grouped by data quality classes.

Quality	Type of SMP signal error	<i>n</i>	%
C1	None	158	60
C2	Trend or offset in absolute SMP force	75	29
C3	Dampened or disturbed SMP force micro-variance	5	2
C4	Both, C2 and C3	24	9

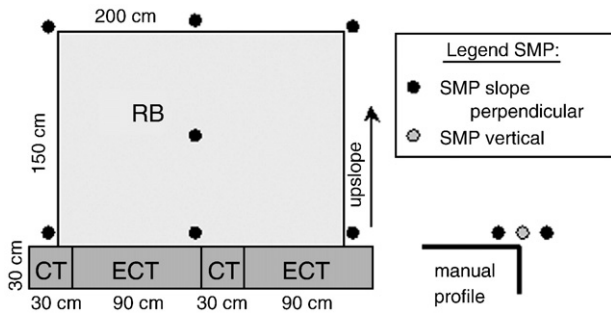


Fig. 3. Experimental design of combined profiles, where one manual profile was taken with three adjacent SMP measurements along with one Rutschblock test, two extended column tests and two compression tests with seven adjacent SMP measurements.

of the combined profiles. The ECT and CT scores are not subject of this study, but are analyzed in a comparison of stability tests by Winkler and Schweizer (2009).

The weak layer depth and the stability were determined by the Rutschblock test (area = 3 m²) as described by Föhn (1987). The RB score (1 to 7), release type (whole block vs. partial break/edge) and fracture surface character (clean vs. rough/irregular) were observed (Schweizer, 2002). Within the immediate vicinity of the manual profile (5 to 20 cm), two slope perpendicular and one vertical SMP measurements were taken. At 29 of 36 sites, six SMP measurements were taken at the perimeter and one in the center of the Rutschblock area (Fig. 3). At the remaining 7 sites, between 2 and 6 SMP measurements were taken at the Rutschblock. The SMP measurements were taken before the Rutschblock was isolated.

3.2. Analysis methods

3.2.1. Stability classification

The snowpack stability for artificial triggering was classified according to the RB scores. We modified the classification by Schweizer et al. (2008) in such that we classified the intermediate stability class (RB score 4 and RB release type whole block) as unstable. Hence, profiles were classified unstable for RB score <4 and RB score 4 with release type whole block ($n = 140$), and stable otherwise ($n = 122$). Only the primary fractures of the Rutschblock test were considered. To objectively classify the stability of the snowpack at the RB scale, we did not consider signs of slope-scale instability such as snowpack settlements (whumpfs) and cracks.

3.2.2. SMP signal analysis

By graphically superimposing the manual profile onto the slope-perpendicular SMP measurement that was closest to the manual profile, the layer boundaries at the failure interface were manually delineated. In cases where the layer matching was ambiguous, the vertical SMP measurement was used as reference. Furthermore, all SMP measurements from one site were graphically aligned to track the failure interface. An example of the manual delineation of the weak layer (WL), the transitional segment (TS), the adjacent layer (AL) and the slab layer (SL) is shown in Fig. 4. The AL can be the stratigraphic layer below the failure interface, as in Fig. 4 or the layer above, depending on where the fracture occurred in the WL.

The SMP signal was first filtered to remove signal noise. Three different methods were applied to the SMP signal after the force normal to tip (F) was calculated: a static threshold of 0.023 N rupture force (based on the 99th percentile of drops in resistance in fluctuations in air measurements (Johnson and Schneebeli, 1999), a dynamic threshold of 10% of the maximum rupture force, and a combination of both. In addition to data quality, the effect of filter type on the results was quantified. Rate-dependent thresholds, as applied by Kronholm (2004), were not compared in this study.

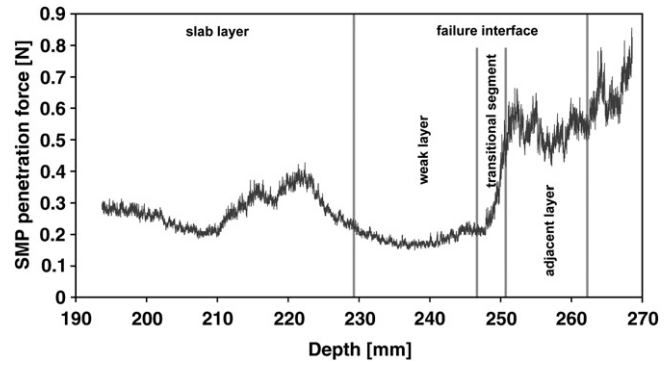


Fig. 4. Section of an SMP profile showing manually depicted layer boundaries at a failure interface.

The calculated SMP parameters based on Marshall and Johnson (2009) are: Rupture force (f), deflection at rupture (δ), structural element length (L), probability of contact (P_c), number of elements engaged (N_e), number of elements available (N_a), mean force (F_m), total force at peak (F_T), stiffness (k), micro-scale elastic modulus (E_{micro}), micro-scale strength (σ_{micro}), measured number of ruptures per mm (N_m) and total number of ruptures per mm (N_T). Also, the texture index (TI) (Schneebeli et al., 1999), the slab layer mean density (ρ) (Pielmeier, 2003), and the depth of the weak layer were calculated. The micro-scale strength is defined as

$$\sigma_{\text{micro}} = \frac{f}{L^2}$$

3.2.3. Statistical analysis

To find the best classifier for unstable and stable RB results from the SMP data, we used the classification tree method (Breiman et al., 1984). The 16 parameters mentioned above were calculated for each SMP profile from 5 mm overlapping windows at 1 mm intervals. Only windows, which were completely within a layer were used. When the layer was less than 5 mm thick, the estimate closest to the middle of the layer was used. The median value of these 16 parameters was calculated for the manually defined weak layer and slab layer as well as the difference between the slab and weak layer. In 95% of all 262 SMP measurements there were at least 6 estimates to derive the median. Together with the weak layer depth, in total 49 variables were tested for their ability to correctly classify the profile based on the RB score. A cross-validation analysis was performed by subdividing the dataset into 10 balanced independent subsets, on which the classifications were performed. This analysis indicated that a 2-node tree, using 2 variables, was statistically significant. We therefore focus below on the classification accuracy using 1 and 2 explanatory variables.

For the analysis of the variance of the best classifier, we used Fisher's one-way ANOVA (Helsel and Hirsch, 2002) on all sites with more than one C1 SMP measurement at the RB perimeter (sites $n = 29$, SMP $n = 123$) as well as a comparison of standard deviations and ranges of the means of weak layer micro-scale strength values at a given site.

3.2.4. Monte-Carlo simulations

To quantify the effect of the number of SMP measurements at a given site on classification accuracy, we performed 10⁴ Monte-Carlo simulations, using boot-strap sampling with replacement (Bevington and Robinson, 1992). Only the sites with at least 7 SMP measurements were used, resulting in 33 of 36 total sites. Quality of the SMP measurements was not taken into account – all measurements were

used in the simulations. One SMP measurement was chosen at random from each site, and the univariate classification was applied. The total, unstable, and stable accuracies were tabulated, and the procedure was repeated 10^4 times. Next, for each site, 2 SMP measurements were chosen at random, and the mean value of the weak layer strength was calculated, and used in the univariate classification. The total, unstable, and stable accuracies were again tabulated, and the procedure was repeated 10^4 times. These Monte-Carlo simulations were repeated, using the average of 3 to 7 SMP measurements. These results are used to quantify the increase in classification accuracy as the number of SMP measurements at a site increases.

4. Results and analyses

4.1. Classification

From the 49 variables used in the univariate classification tree analysis, the ten best predictors of unstable and stable profiles are given in Table 2.

The single best predictor was the weak layer micro-scale strength ($WL_{\sigma_{micro}}$), with values $\geq 0.0755 \text{ N mm}^{-2}$ classified as stable, and $< 0.0755 \text{ N mm}^{-2}$ as unstable, with a total classification accuracy of 83.5% (stable accuracy 79.5%, unstable accuracy 87.5%). When all 262 SMP profiles were grouped by unstable and stable Rutschblock scores, the distributions of $WL_{\sigma_{micro}}$ were significantly different.

From the multivariate classification tree analysis of the 49 variables, the second-best variable for predicting unstable and stable profiles (with $WL_{\sigma_{micro}}$ as first variable) is the slab layer mean density (SL_{ρ}). The second-best variables with the 12 highest total accuracies are given in Table 3. Also, the total accuracies in terms of the data quality classes are shown.

From the classification tree analysis shown in Fig. 5, the SMP failure interfaces were predicted to be stable if $WL_{\sigma_{micro}} \geq 0.0755 \text{ N/mm}^2$ and $SL_{\rho} \leq 320.6 \text{ kg/m}^3$, and unstable otherwise.

Fig. 6 shows all data along with the split values gained from the classification tree analysis, as a function of data quality class. The total classification accuracy for all data was 85.5%.

4.2. Sensitivity analysis

4.2.1. Sensitivity to SMP data quality

For the univariate classification, with $WL_{\sigma_{micro}}$ as best classifier (Table 2), the accuracies were calculated for three different SMP data quality classes. Table 4 shows that there were lower accuracies for C2 data than for C1 data, and even lower accuracies for C3 and C4 data.

For the multivariate classification, with $WL_{\sigma_{micro}}$ and SL_{ρ} as best classifiers (Table 3), the accuracies were less sensitive to data quality, as shown in Table 5.

Table 2

The ten best classifiers from univariate classification tree analysis. TA is the total accuracy, SA is the stable accuracy, UA is the unstable accuracy. At the split value of the SMP parameter the classification changes from stable to unstable or vice versa. A number one in the “ $\geq$ ” column means the split was greater or equal than the value to be stable, a zero means less than the split value is stable.

SMP parameter	TA [%]	SA [%]	UA [%]	Split value	$\geq$
$WL_{\sigma_{micro}}$	83.5	79.5	87.5	0.0755 N/mm ²	1
SL_{ρ}	79.8	89.7	70.0	0.13 [prob.]	0
WL_{N_e}	77.2	73.1	81.3	14 [#]	1
WL_{N_a}	77.2	73.1	81.3	33 [#]	1
$WL_{\sigma_{micro}}$	77.2	92.3	62.5	1.44 N/mm ²	1
WL_L	77.2	73.1	81.3	1.09 mm	0
WL_{N_T}	76.6	69.2	83.8	82 [#]	1
WL_F	75.3	65.4	85.0	0.10 N	1
WL_k	74.1	87.2	61.3	1.45 N/mm	1
WL_{depth}	73.4	74.4	72.5	200 mm	1

Table 3

Total accuracies obtained from the multivariate tree analysis (with $WL_{\sigma_{micro}}$ as first variable). The total accuracy is shown for the twelve best second variables, also as a function of SMP data quality classes.

SMP parameter	C1 [%]	C2 [%]	C3,C4 [%]	C1,C2 [%]	C1,C2, C3,C4 [%]
SL_{ρ}	85.44	89.33	75.86	88.00	85.50
SL_{TI}	84.18	85.33	68.97	84.55	82.82
SL_F	85.44	81.33	65.52	84.12	82.06
WL_L	84.18	84.00	82.76	84.12	83.97
WL_{N_a}	83.54	82.67	82.76	83.26	83.21
$WL_{\sigma_{micro}}$	84.81	80.00	79.31	83.26	82.82
WL_{Nm}	81.01	81.33	82.76	81.12	81.30
WL_{N_T}	81.65	80.00	82.76	81.12	81.30
WL_{P_c}	82.28	76.00	72.41	80.26	79.39
WL_{F_m}	77.85	84.00	72.41	79.83	79.01
SL_{P_c}	84.81	69.33	68.97	79.83	78.63
WL_{depth}	83.54	70.67	72.41	79.40	78.63

The multivariate classification was robust to C2 data (29% of all data), where the signal error was a trend or an offset in absolute SMP force. C1 and C2 data together yielded a total accuracy of 88%. C3 and C4 data significantly reduced the total classification accuracy by misclassifying more than half of the stable cases. Reduced micro-variance leads to more unstable classifications because the rupture force values are reduced which in turn leads to reduced values of σ_{micro} . However, the accuracies obtained using all data together were still acceptable (total accuracy = 85.5%), indicating that this method may be applicable to large datasets not manually classified by quality, which may be too time consuming to do in some cases.

4.2.2. Sensitivity to SMP noise removal methods

The classification accuracies for $WL_{\sigma_{micro}}$ (Table 2) were calculated for three different SMP signal noise removal methods: a static filter, a dynamic filter and a combination of both. Table 6 shows that the classification accuracies were not sensitive to the selected filter method.

4.2.3. Sensitivity to weak layer thickness

The vast majority of the 262 SMP measurements (95%) contain weak layers that are thicker than 0.5 cm. In 65% of the cases, the weak layer is thicker than 3 cm. To show the effect of weak layer thickness the dataset was subdivided in weak layers of 3 cm and thinner and weak layers thicker than 3 cm. Fig. 7 shows that the classification is robust to weak layer thickness.

4.3. Analysis of variance

The univariate analysis of variance (one-way ANOVA) of $WL_{\sigma_{micro}}$ showed that the measurements were significantly different (p -values < 0.05) within any given site. This means that, even in situations with low RB scores and whole block release types, at least one SMP measurement within a RB area had a $WL_{\sigma_{micro}}$ that was significantly different from the other ones. Table 7 shows the standard deviations and ranges of the means of $WL_{\sigma_{micro}}$ for stable and unstable sites, for C1 data. These results show that the variance of the $WL_{\sigma_{micro}}$ mean values per site was not always smaller in unstable than in stable situations.

4.4. Effect of the number of available SMP measurements at a site

The dataset presented in this paper is unique, as multiple SMP measurements were made at each site where a RB test was performed. We hypothesized that increasing the number of SMP measurements near the RB would improve the accuracy with which the RB stability could be predicted. The distribution of weak layer micro-scale strength for all 262 SMP measurements, separated by RB stability shows, that during unstable conditions, the weak layer micro-scale strength is generally lower than during stable conditions and that the

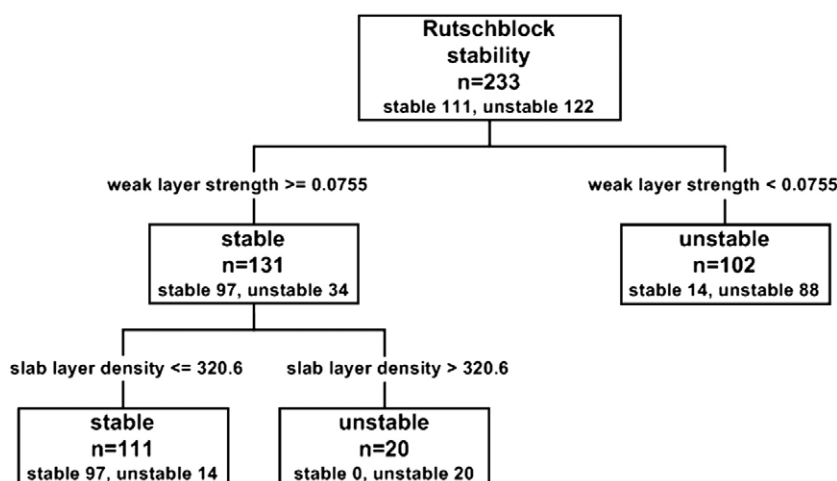


Fig. 5. Classification tree for unstable/stable dataset ($n = 233$, C1 and C2 data). The cross-validation analysis indicated that the two-node tree, using two variables is statistically significant.

IQR for the stable and unstable conditions do not overlap (Fig. 8). The distributions are significantly different, and it is clear that reasonable classification accuracy can be achieved using $WL_{\sigma_{\text{micro}}}$.

In the right panel in Fig. 8, the distribution of the mean $WL_{\sigma_{\text{micro}}}$, calculated using all SMP profiles at a given site, is shown. These distributions are more clearly separated and the variance is reduced, indicating that using a mean value from multiple SMP measurements improves the classification accuracy. The split value used in the above analysis is shown with the black dashed line on the left panel. When mean values are used, the optimal split value is 0.1 N mm^{-2} , shown as a black dashed line in the right panel. But how many SMP measurements are required to calculate an accurate mean value at the RB scale?

The results of the Monte-Carlo simulations, which were repeated for a mean value calculated from one to seven SMP measurements, using a split value of 0.1 N mm^{-2} shows, that the accuracy increases with the number of SMP measurements available for the estimate of the mean (Fig. 9). When seven SMP measurements were available to estimate the mean weak layer micro-scale strength, 95% of the time the total classification accuracy was at least 85%.

5. Discussion and conclusion

Applying the improved SMP signal interpretation theory by Marshall and Johnson (2009) to a dataset that combines snow profiles, multiple SMP measurements and Rutschblock tests, significant indicators of snowpack instabilities were derived from SMP measurements. $WL_{\sigma_{\text{micro}}}$ was the single most significant parameter, whereby a split value of 0.1 N mm^{-2} effectively separated nearly all unstable from stable observations. Lutz et al. (2009) identified the same optimum split-value for discerning stable from unstable conditions, despite applying a different experimental design. Combining $WL_{\sigma_{\text{micro}}}$ with the second-best parameter from multivariate analysis, SL_{ρ} , the total classification accuracy for C1 and C2 data was 88% where false alarms and misses are balanced (stable accuracy 87.4%, unstable accuracy 88.5%).

The presented classification model is a considerable improvement to previous studies and can serve as a basis for development of an automated detection of potential instabilities in snowpacks from SMP measurements.

For the first time, SMP data quality and a selection of signal filter methods were controlled in a classification analysis. The sensitivity studies for SMP data quality showed that the classification model is

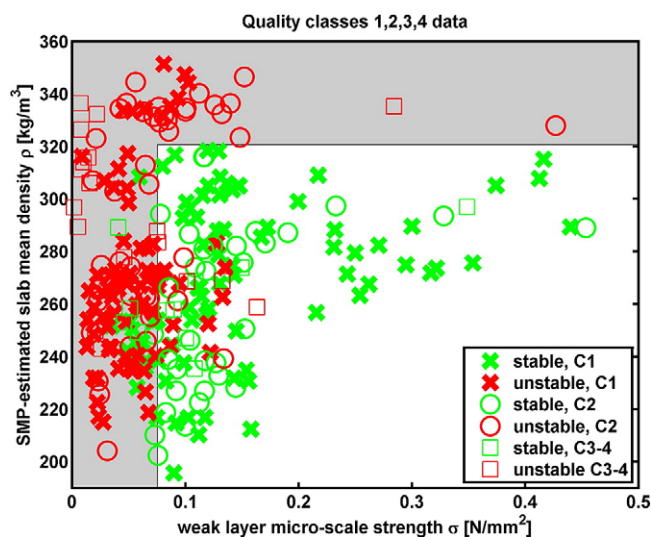


Fig. 6. Plot values of weak layer micro-scale strength as a function of slab layer mean density for all SMP data quality classes. The grey shaded area is the two variable best classification for Rutschblock stability class. The different symbols indicate SMP data quality classes.

Table 4

The classification accuracies for weak layer micro-scale strength in terms of SMP data quality classes. TA is the total accuracy, SA is the stable accuracy, UA is the unstable accuracy, and n is the number of SMP profiles in the group.

Classification accuracies	C1	C2	C3,C4
TA [%]	83.54	74.67	68.97
SA [%]	79.49	78.79	72.73
UA [%]	87.50	71.43	66.67
n	158	75	29

Table 5

The classification accuracies for combined weak layer micro-scale strength and slab layer mean density in terms of SMP data quality classes. TA is the total accuracy, SA is the stable accuracy and UA is the unstable accuracy.

Classification accuracies	C1	C2	C3, C4	C1, C2	C1,C2, C3,C4
TA [%]	85.44	89.33	75.86	88.00	85.50
SA [%]	79.49	87.88	45.46	87.40	78.12
UA [%]	91.25	90.48	94.44	88.50	91.38

Table 6

The classification accuracies for weak layer micro-scale strength from univariate analysis in terms of SMP noise removal methods for C1 data.

Classification accuracies	Static	Dynamic	Static, dynamic
TA [%]	83.54	84.81	83.54
SA [%]	79.49	78.49	79.49
UA [%]	87.50	90.00	87.50
<i>n</i>	158	158	158

robust to trends and offsets in the SMP mean force (i.e., C2 data). In this dataset, this was by far the most frequent signal error encountered (29% of all data, Table 1). But, the classification is sensitive to a dampened SMP force micro variance (C3 and C4 data), where particularly unstable profiles are misclassified. In our study, only 11% of the data was C3 or C4. If this proportion were larger, the classification accuracy without quality control would likely be less. For any study with SMP data it is important to determine whether the analysis method is sensitive to errors in the mean force, in the variance of the force or both. If high accuracy is required, the consideration of data quality will improve the results. SMP data quality control is also recommended, when datasets acquired with different SMP probes are merged.

Furthermore, the sensitivity analysis showed that the best classifier, $WL_{\sigma_{micro}}$, is robust to the choice of SMP signal noise removal methods. Therefore we suggest that SMP users apply the simplest method, the static filter with a rupture force threshold value based on fluctuations in air signals, and focus on the strength parameter for stability applications. For our dataset, layer thickness does not affect the classification.

This study demonstrates the improvement in classification accuracy when multiple SMP measurements are made at a given site. It is based on profiles with a wide range of snowpack types and weak layers. However, the dataset contains only few very thin weak layers and it does not contain buried surface hoar layers. When only one SMP measurement is available at the Rutschblock scale, the total classification accuracy can vary from 60% to 95%, due to the difference in sampled area between the SMP (1 cm²) and the RB (20,000 cm²). As the number of SMP measurements available for estimating the mean weak layer micro-scale strength increases, the accuracy increases, at the cost of sampling time. Even when four SMP measurements are available, there is a large range in total accuracy.

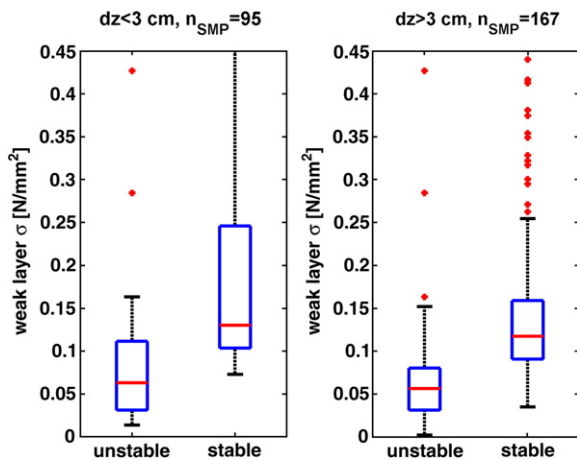


Fig. 7. Distribution of $WL_{\sigma_{micro}}$ for stable and unstable RB stability classes. Left panel shows results for weak layers 3 cm and thinner, and right panel shows the results for weak layers thicker than 3 cm. The weak layer thickness (*dz*) has very little effect on the result. The box represents the middle 50% of the data (IQR), the line within the box is the median value, and the whiskers represent the most extreme values within 1.5 times the IQR from the ends of the box. Outliers beyond the ends of the whiskers are shown with “+” symbols.

Table 7

Standard deviations (std) and ranges of the $WL_{\sigma_{micro}}$ means per site according to stability class. *n* is the number of SMP measurements per given site with two and more SMP measurements at the RB and C1 data. The data is in ascending order of the standard deviation of the means at a given site.

Stable site no.	Std of means	Range of means	<i>n</i>	Unstable site no.	Std of means	Range of means	<i>n</i>
33	0.0016	0.0023	2	9	0.0026	0.0049	3
24	0.0053	0.0151	6	16	0.0050	0.0131	7
5	0.0104	0.0261	5	13	0.0053	0.0115	5
23	0.0140	0.0432	6	25	0.0061	0.0140	6
6	0.0203	0.0399	3	10	0.0066	0.0145	4
8	0.0232	0.0544	4	36	0.0076	0.0221	6
32	0.0303	0.0783	5	12	0.0129	0.0273	4
34	0.0339	0.0983	6	14	0.0143	0.0300	4
20	0.0593	0.1583	6	35	0.0153	0.0358	4
7	0.0655	0.1515	4	3	0.0171	0.0430	6
26	0.0957	0.2757	6	21	0.0280	0.0655	6
2	0.1272	0.3075	7	11	0.0299	0.0721	5
27	0.1369	0.1936	2				
30	0.1661	0.3304	3				

When seven SMP measurements are available to estimate the mean, the total accuracy ranged from 82 to 87%, the unstable accuracy was always at least 87%, and the stable accuracy was 75 to 82%. In 95% of the simulations, using one SMP measurement gave a classification accuracy of at least 69%. When seven SMP measurements were available to estimate the mean $WL_{\sigma_{micro}}$, 95% of the time the total classification accuracy was at least 85%. Since SMP measurements are relatively fast, it is generally beneficial to take multiple measurements. Estimating snowpack stability at the Rutschblock-scale using a single SMP measurement holds a high risk of being too far from the mean for this type of analyses.

Future work involves the consideration of the bed-surface layer in the classification. An extension of dataset to profiles with buried surface hoar layers would round off the model. A catalogue of data quality classes and the development of an algorithm to detect errors in the SMP force would greatly assist SMP users. With an automatic weak layer detection procedure from multiple SMP measurements, the SMP has the potential to become useful in operational snowpack assessments. The study provides a basis for further developments in the study of slope-scale snowpack stability.

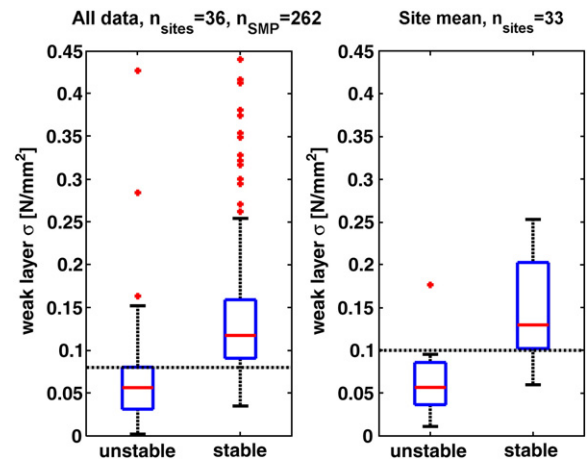


Fig. 8. Distribution of $WL_{\sigma_{micro}}$ for stable and unstable RB stability classes. Left panel shows results from all 262 SMP profiles, and right panel shows the mean of the weak layer strength at each site where there were at least seven measurements. Using the mean of multiple SMP measurements at a site reduces the variance of the results and hence improves the classification accuracy. The box represents the middle 50% of the data (IQR), the line within the box is the median value, and the whiskers represent the most extreme values within 1.5 times the IQR from the ends of the box. Outliers beyond the ends of the whiskers are shown with “+” symbols.

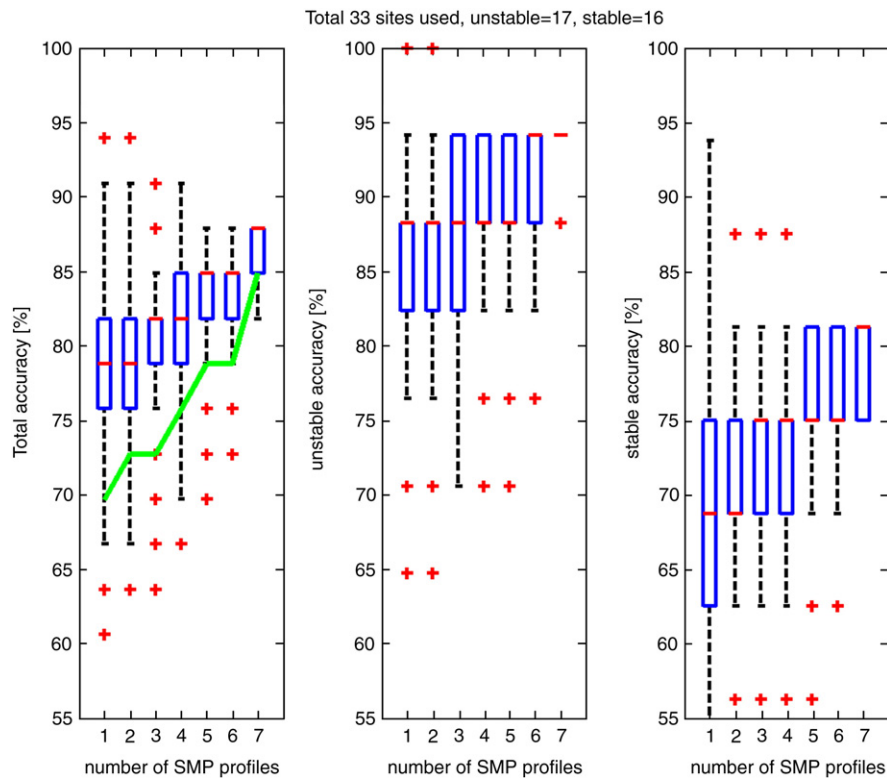


Fig. 9. Classification accuracies as a function of the number of SMP measurements used to estimate mean $WL_{\sigma_{micro}}$ at a site, from 10^4 Monte-Carlo simulations, including total accuracy (left panel), unstable accuracy (middle panel), and stable accuracy (right panel). There is a steep increase in accuracy with the increasing number of SMP measurements available for the estimate of the mean. The curve line (left panel) shows the 5% quantile; 95% of the time, the accuracy was above this value. The box represents the middle 50% of the data (IQR), the line within the box is the median value, and the whiskers represent the most extreme values within 1.5 times the IQR from the ends of the box. Outliers beyond the ends of the whiskers are shown with “+” symbols.

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